

Kaiwen Hong

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<https://kaiwen-hong.github.io/>

Research focus: adaptable, deployable, and steerable robots — VLAs, world models, long-horizon manipulation.

EDUCATION

- University of Illinois at Urbana-Champaign** Sept. 2021 - Aug. 2026 (Expected)
Ph.D. in ECE, advised by Prof. Katherine Driggs-Campbell GPA: 3.84/4.0
- University of Illinois at Urbana-Champaign** Aug. 2017 - May 2021
B.S. in Electrical Engineering (dual degree), High Honors GPA: 3.87/4.0
- Zhejiang University** Aug. 2017 - May 2021
B.Eng. in EE and Automation (dual degree), Outstanding Graduate GPA: 3.95/4.0

RESEARCH EXPERIENCE

- Self-Labelled Preference Transfer (SPT) for Steerable Manipulation** Jan. 2026 - May 2026
 - Proposed **SPT**, a two-stage VLA framework (built on [StarVLA](#)) that removes per-task preference annotation: co-train a preference-conditioned action head with a preference-recognition head on labeled source tasks, then self-label unlabeled target-task demonstrations to transfer preference-following to new tasks.
 - Achieved **96.4%** preference-recognition accuracy and **+25.4%** preference-following over the naive VLA baseline in simulation and challenging real-world tasks (data via [FastUMI](#); wrist-fisheye camera deployment).
- MemCompiler: Agent-Compiled Memory for Long-Horizon Manipulation** Jan. 2026 - Present
 - Designing an **agentic memory system** where a VLM agent prospectively writes explicit, verifiable memory (SAM2 object tracks, event logs, symbolic counters) and compiles it into spatial conditions for a **frozen** $\pi_{0.5}$ policy: **2–3 $\times$** success on the hardest object-permanence tasks (e.g., **42% $\rightarrow$ 90%**) with zero policy training.
- DiscreteRTC: Discrete Diffusion Policies as Asynchronous Executors** Jan. 2026 - May 2026
 - Co-developed **DiscreteRTC**, showing discrete diffusion policies are natural asynchronous executors for latency-critical dynamic tasks; **+50%** success over Real-Time Chunking (**RTC**) on real-world tasks. [\[website\]](#)
- Sim-and-Real Co-Training of Graph Neural Dynamics for Scooping** May 2023 - Sept. 2025
 - Developed a **sim-and-real co-training** framework for long-horizon granular scooping: a graph-neural **world model** trained jointly on large-scale simulation and a small real-world dataset, with **Monte-Carlo tree search** planning over multiview calibrated RGB-D point clouds (**ICRA 2026**). [\[website\]](#)

INDUSTRIAL RESEARCH EXPERIENCE

- Waymo LLC** June 2026 - Aug. 2026
 - Research Intern, Mountain View, CA*
 - Researching a **generalized renoising policy** for **uniform-state discrete diffusion (USD)** VLMs (beyond masked diffusion), built on [DiffusionGemma](#) for long-tail autonomous-driving problems.
 - Post-training on driving data with SFT and RL-based preference alignment, targeting quality that matches or exceeds autoregressive VLMs while keeping parallel-decoding efficiency (ongoing).
- Amazon Robotics** May 2025 - Dec. 2025
 - Applied Scientist II Intern, Seattle, WA*
 - Delivered a transformer-based progress-prediction model (on DINOv2 features) for [Vulcan Stow](#)—the first to beat the expert heuristic (**+24% F1**); deployed to production across ~ 10 robots nationwide, driving abort decisions, affordance planning, and throughput (UPH).
 - Post-trained a reasoning-augmented VLM (SFT on [Qwen2.5-VL](#)) for scene-to-affordance prediction on long-tail cases, building a chain-of-thought reasoning dataset from scratch; **+12%** affordance accuracy over the production baseline.
- XPENG Inc.** May 2024 - May 2025
 - ML Research Intern, San Diego, CA*
 - Led development of a driving **world model** (VQ-VAE tokenizer + autoregressive Transformer, [GAIA-1](#)-style) on 2M ~ 30 s video clips; used for downstream planning, **+15% Driving Score** over SOTA on [Bench2Drive](#).
 - Trained a cross-unified embodiment VLA based on Qwen2.5-VL jointly on [Open X-Embodiment](#) and internal autonomous-driving data, beating [UniAct](#) by **+10%** on [LIBERO](#) and reaching SOTA (**82.5**) on [NavSim](#).

SELECTED PUBLICATIONS

* Equal contribution.

- **K. Hong**, H. Chen*, J. Xu*, R. Wang*, K. Wang*, M. Zhang, S. Liu, Y. Zhu, Y. Li, and K. Driggs-Campbell, “Sim-and-Real Co-Training of Graph Neural Dynamics for Long-Horizon Scooping,” in **ICRA 2026**. [\[website\]](#)
- **K. Hong**, S. Peng*, X. Zeng*, P. Wang*, B. Tian, H. Chen, Y. Zhu, Y. Wang, and K. Driggs-Campbell, “Self-Labeled Preference Transfer: Bootstrapping Steerable Robot Manipulation on Unlabeled Tasks,” in submission.
- **K. Hong**, P. Wang, H. Jiang, X. Zeng, S. Peng, B. Tian, H. Chen, Y. Zhu, L. Sun, and K. Driggs-Campbell, “MemCompiler: Agent-Compiled Memory for Long-Horizon Manipulation,” in preparation.
- P. Wang, **K. Hong**, C. Peng, K. Driggs-Campbell, M. Tomizuka, C. Xu, and C. Tang, “DiscreteRTC: Discrete Diffusion Policies are Natural Asynchronous Executors,” in submission. [\[website\]](#)
- H. Chen, **K. Hong***, Y. Niu*, S. Liu, Y. Wang, Y. Li, and K. Driggs-Campbell, “Predicting Object Interactions with Behavior Primitives: An Application in Stowing Tasks,” in **CoRL 2023 (Oral, Best Paper / Best Student Paper Award Finalist)**. [\[website\]](#)
- S. Luo*, Q. Peng*, J. Lv*, **K. Hong**, K. Driggs-Campbell, C. Lu, and Y. Li, “Human-Agent Joint Learning for Efficient Robot Manipulation Skill Acquisition,” in **ICRA 2025 (Best Paper in Human-Robot Interaction and Best Conference Paper Finalist)**. [\[website\]](#)
- S. Liu, P. Chang, Z. Huang, N. Chakraborty, **K. Hong**, W. Liang, J. Geng, and K. Driggs-Campbell, “Intention-Aware Robot Crowd Navigation with Attention-Based Interaction Graph,” in **ICRA 2023 (Best Poster Award at IROS 2023 Workshop)**. [\[website\]](#)
- S. Liu, H. Xia, F. Pouria, **K. Hong**, N. Chakraborty, Z. Hu, J. Biswas, and K. Driggs-Campbell, “HEIGHT: Heterogeneous Interaction Graph Transformer for Robot Navigation in Crowded and Constrained Environments,” in **IEEE T-ASE (Best Paper Award at ICRA 2025 Workshop)**. [\[website\]](#)
- H. Chen, J. Xu*, H. Chen*, **K. Hong**, B. Huang, C. Liu, J. Mao, Y. Li, Y. Du, and K. Driggs-Campbell, “Multi-Modal Manipulation via Multi-Modal Policy Consensus,” in **ICRA 2026 (Best Paper Award at CVPR 2026 Workshop)**. [\[website\]](#)

SELECTED AWARDS

- **Best Paper / Best Student Paper Award Finalist, CoRL 2023**
- **Best Conference Paper Finalist, ICRA 2025**
- **Best Paper in Human-Robot Interaction, ICRA 2025**
- **Best Paper Award, CVPR 2026 Sense of Space Workshop**
- **Best Paper Award, ICRA 2025 Workshop on Advances in Social Navigation**
- **Best Poster Award, IROS 2023 Last-Mile Robotics Workshop**
- **Second Prize, RoboMaster International Region Competition (ZJU-UIUC Team), 2019**

OTHER PROJECTS & EXPERIENCE

- **Stealth AI Startup — Research Intern (Remote)** Jan. 2026 - May 2026
 - Post-trained a visual-tactile **world model** on [Cosmos-Predict 2.5](#), validated in simulation ([UniVTAC](#)) and on an internal humanoid with a dexterous hand; reached **90%** on real-world peg insertion using a large-scale dataset collected via [Sunday Robotics](#)-style gloves.
- **Predicting Object Interactions with Behavior Primitives: Stowing** Oct. 2022 - May 2023
 - Combined **GNN-based object-interaction prediction** with primitive-augmented trajectory optimization to stow objects into cluttered shelves from a single demonstration (3–4 keyframes); **95%** real-world success. **Best Paper / Best Student Paper Award Finalist, CoRL 2023**. [\[website\]](#)
- **Crowd-Aware Social Navigation** Jan. 2022 - Sept. 2025
 - Built attention-based recurrent GNNs that fuse heterogeneous spatio-temporal interactions with RL and trajectory prediction; deployed on a TurtleBot 2i (**ICRA 2023**) and extended to crowded, constrained environments with **HEIGHT (T-ASE)**; best paper/poster awards at ICRA 2025 and IROS 2023 workshops.

TECHNICAL SKILLS

- **ML / DL:** PyTorch, VLAs ($\pi_{0.5}$, StarVLA), VLMs (Gemma, QwenVL, DreamVL), Video Models (WAN, Cosmos-predict), GNNs, Imitation Learning, Reinforcement Learning (PPO, GRPO)
- **Robotics:** ROS, real-robot deployment (UR5e, Kinova, Franka, Stretch, TurtleBot 2i), data collection & deployment of imitation learning pipeline (FastUMI, SpaceMouse, UMI gloves), simulation (Isaac Sim, MuJoCo, Sapien, Genesis)